Computer Vision 2026: Assignment 04

Michael Bezuidenhout [1]

1. Image Retrieval for Image Re-Identification

1.1. Problem Statement and Approach

A small image embedding model was trained on the VeRi dataset (Liu et al., 2016a) for use in a vehicle re- identification system. The aim was to train a model that is capable of producing an embedding space where images of the same vehicle are clustered relatively close to each other, while images of different vehicles are pushed farther apart. During evaluation, query and gallery images are converted to embedded feature vectors and each query is matched to gallery images in order of proximity within the embedding space. A capable model should be able to rank gallery images of the particular vehicle above vehicles of similar colour and shape (not to mention vehicles that are completely different).

1.2. Dataset

The VeRi dataset (Liu et al., 2016a) is a large-scale dataset for vehicle re-identification and was used with permission for this assignment. The dataset contains over 50 000 images of 776 vehicles captured by 20 cameras in a real-world traffic setting and is split into training, query, and test (gallery) sets. The re-identification task is to match query images to gallery images based on the vehicle ID using an embedding model that is trained on the training set.

Also included in the dataset are detailed annotations for each image, including the vehicle ID, camera ID, and bounding box coordinates. For this assignment, only the vehicle ID and images were used for training and evaluation. Image labels were extracted directly from the file name.

Image preprocessing consisted of resizing all images to a fixed resolution of 128×128 pixels, converting the images to PyTorch tensors, and normalizing to the mean and standard deviation of the ImageNet dataset to match the pre-trained ResNet backbone. Constraining the image resolution was a necessary compromise to keep the training requirement within reasonable limits for the scope of this assignment. It is recommended that higher image resolutions are investigated in future work.

1.3. Model Architecture

The model architecture can be compartmentalised into two functional components: the backbone for feature extraction and the embedding head for producing the final feature space embedding.

The backbone is comprised of a pre-trained ResNet-34 (He et al., 2015) model with the final fully connected classification layer removed since only the feature extraction capabilities are needed. This leaves only the residual convolutional blocks of the ResNet model. After global average pooling, the output from the convolutional feature extractor is a 512-dimensional feature vector.

The ResNet backbone was initialized with pre-trained weights from training on the ImageNet dataset. The weights were frozen during training on the VeRi dataset to save compute.

The embedding head is a simple feedforward network with a single hidden layer of 256 units and SiLU activation, followed by a final output layer of 256 units. The final output is not passed through an activation function but is instead L2-normalized to produce the final embedding. The benefit of normalization is that the embedding space and vector comparison is not influenced by the magnitude of feature vectors, thereby stabilizing the comparison and improving consistency.

The decision to use ResNet34 as a backbone was ultimately made as a compromise between the computational resources and model capacity. A deeper ResNet model would likely produce better results but the resource requirement proved too significant for the scope of this assignment. It is recommended that a deeper architecture and higher image resolution is investigated in future work to potentially improve performance.

1.4. Triplet Mining Strategy

The model was trained using a triplet mining strategy with semi-hard negatives. A triplet consists of an anchor image (a), a positive image (p) and a negative image (n). The positive image is of the same class (vehicle ID) as the anchor, while the negative is of a different class. The aim is to train the model such that the distance between the anchor and positive is less than the distance between the anchor and negative by at least a margin α.

Mini-batches were constructed for training by using a P $\times$ K strategy, where P is the number of classes (vehicle IDs) and K is the number of images per class (Brink, 2026). A batch size of 32 was used with P = 8 and K = 4 in an attempt to balance computational requirements and triplet diversity.

Semi-hard negative mining was then implemented within each batch. Semi-hard negatives are those that are farther from the anchor than the positive but still within the margin α:

$$D_{ap} < D_{an} < D_{ap} + \alpha$$

This strategy is valuable because it moderates the difficulty of training examples. Negative samples which are already located far from the anchor are not as valuable for learning as those which are closer to the anchor and thus more likely to be confused with the positive. It is also a practical consideration since compute is not wasted on extreme negatives which are unlikely to be informative for training.

Since the model outputs L2-normalized embeddings, the maximum Euclidean distance between two points in the embedding space is 2. This effectively also constrains the margin α to a sensibly smaller value within this range. Since the optimal margin is not known, values of α between 0.2 and 0.5 were investigated.

Random triplet sampling was also investigated but proved to be less effective than semi-hard negative sampling with the P $\times$ K batch construction while also slowing training due to the need to mine for triplets across the entire dataset.

1.5. Training

After a broad initial investigation, the final set of models were trained using the PK batch sampling strategy with semi-hard negative mining and varied margins α of 0.2 to 0.5. The mini-batch size was 32 ($P = 8, K = 4$) to balance resource requirements and triplet diversity. The AdamW optimizer was used with a learning rate of 1e-3 and a weight decay of 1e-4. Model configurations were trained for 50 epochs each.

Triplet loss was calculated using a hinge component to ensure that only triplets which violate the margin contribute to the loss. The loss for a single triplet is calculated as follows:

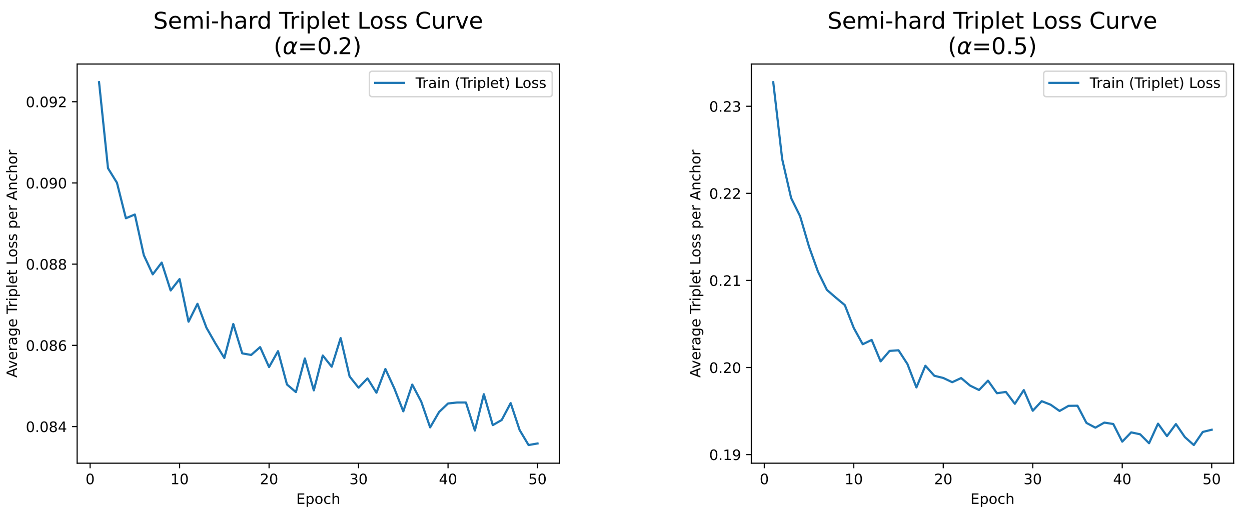
$$L = \max(0, D_{ap} - D_{an} + \alpha) \tag{1}$$

where α is the margin and D_{ap} and D_{an} are the Euclidean distances between the anchor and positive, and anchor and negative respectively. The total loss for an epoch is averaged across all triplets.

Figure 1a and Figure 1b show the triplet loss curve for training with $\alpha = 0.2$ and $\alpha = 0.5$. The loss curves decrease sharply in the early epochs before trending toward a plateau. The loss curve in Figure 1a appears to be more unstable than the curve in Figure 1b, which may be due to the smaller margin resulting in fewer triplets that meet the semi-hard negative criteria and thus more variability in the average loss. Potential improvements to this might have been to decrease the learning rate for the smaller margin to stabilize training, or to increase the batch sampling size to increase the number of triplets that meet the semi-hard negative criteria.

1.6. Results and Discussion

Table 1 shows the retrieval performance of the trained models for different margin values α. The model trained with $\alpha = 0.2$ achieved the best performance across all metrics, with a Top-1 accuracy of 0.9774, Top-5 accuracy of 0.9923, and mAP of 0.1861. As the margin increased, performance generally decreased, with the model trained with $\alpha = 0.5$ achieving the



(a) (b)

Figure 1. Training loss curves for models trained with (a) $\alpha = 0.2$ and (b) $\alpha = 0.5$.

lowest performance with a Top-1 accuracy of 0.8892, Top-5 accuracy of 0.9565, and mAP of 0.1509. This suggests that a smaller margin is more effective for the task of image re-identification in this context, likely because it allows for more informative triplets to contribute to the loss during training.

Table 1. Retrieval performance for different triplet loss margin values.

Margin α	Top-1 Accuracy	Top-5 Accuracy	mAP
0.2	0.9774	0.9923	0.1861
0.3	0.9487	0.9821	0.1771
0.4	0.9231	0.9744	0.1636
0.5	0.8892	0.9565	0.1509

Figure 2 shows the top 5 gallery images retrieved for selected queries using the model trained with $\alpha = 0.2$. In all examples, the top-ranked gallery image is a correct match (same vehicle ID as the query), which is reflective of the high Top-1 accuracy reported in Table 1. In some cases, the lower ranked images are of an incorrect vehicle ID but with similar shape and colour. At first glance, these mismatches look fairly similar to the query image, only differing in some minor details like window stickers. This suggests that the model is learning a reasonably effective embedding space where similar vehicles are clustered together, even if they are not the same vehicle ID.

A deeper convolutional network and higher image resolution may have allowed the model to learn a more detailed embedding space, which in turn may have improved the model's ability to distinguish between similar vehicles.

An observation from Figure 2 is that the top-ranked gallery image looks nearly identical to the query image. This is not ideal because matching near-duplicate images is not representative of the requirements of a real-world re-identification system. The retrieval metrics in Table 1 are likely inflated by the presence of near-duplicate images and cast doubt on the model's ability to replicate the performance in a real-world setting.

An improvement to this shortcoming may be to exclude images from the same camera as the query image from the gallery during evaluation. This would reduce the likelihood of near-duplicate images appearing in the gallery because images from different cameras are more likely to have different angles, lighting conditions, and other variations that make the re-identification task more realistic.

Figure 2. Top 5 gallery images retrieved for selected query images using embeddings produced by the $\alpha = 0.2$ model. Green boxes indicate correct matches (same vehicle ID) and red boxes indicate incorrect matches.

1.7. Recommendations

While the trained models in this assignment served as a good proof-of-concept for the use of triplet loss and semi-hard negative mining for image re-identification, there are several avenues for improvement that could be explored in future work.

Perhaps most notably, future work should focus on implementing a more realistic evaluation protocol. This may involve excluding near-duplicate images from the gallery during evaluation. Further improvements might include investigating different image resolutions, triplet mining strategies, and feature space embeddings. As a final recommendation, extended training time should be investigated to allow the model to converge fully with early stopping to prevent overfitting.

References

Brink, W. Computer vision 842 lecture notes, 2026.

Elgendy, M. *Deep Learning for Vision Systems*. Manning Publications, 2020.

He, K., Zhang, X., Ren, S., and Sun, J. Deep residual learning for image recognition. *Microsoft Research*, 2015.

Liu, X., Liu, W., Ma, H., and Fu, H. Large-scale vehicle re-identification in urban surveillance videos. *ICME*, 2016a.

Liu, X., Liu, W., Mei, T., and Ma, H. A deep learning-based approach to progressive vehicle re-identification for urban surveillance. *ECCV (2)*, 2016b.

Liu, X., Liu, W., Mei, T., and Ma, H. Provid: Progressive and multimodal vehicle re-identification for large scale urban surveillance. *IEEE Transactions on Image Processing*, 2018.

Torralba, A., Isola, P., and Freeman, W. T. *Foundations of Computer Vision*. MIT Press, 2024.